

Yelp Business Analytics API 1.0.0 OAS 3.1

</openapi.json>

API for analyzing business performance with visual analytics

businesses

GET

</api/businesses/> List Businesses

Get all businesses in region with optional filtering.

Supports filtering by state and city for dashboard and initial map view. Returns a paginated list of businesses ordered by star rating and review count.

Parameters

[Try it out](#)

Name	Description
------	-------------

state	State code (e.g., 'PA', 'CA')
-------	-------------------------------

string (string null) (query)	
---	--

minLength: 2
maxLength: 2

city	City name
------	-----------

string (string null) (query)	
---	--

minLength: 1

skip	Number of records to skip
------	---------------------------

integer (query)	
--------------------	--

Default value : 0

minimum: 0

limit	Maximum number of records to return
-------	-------------------------------------

integer (query)	
--------------------	--

Default value : 100

maximum: 1000
minimum: 1

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema [{ "attributes": { "Alcohol": "u'none", "RestaurantsDelivery": "False", "WiFi": "u'free" }, "business_id": "MTSW4McQd7CbVtyjqoe9mw", "categories": "Restaurants, Food, Bubble Tea, Coffee & Tea, Bakeries", "city": "Philadelphia", "hours": { "Monday": "7:0-20:0", "Tuesday": "7:0-20:0" }, "is_open": 1, "latitude": 39.9555052, "longitude": -75.1555641, "name": "St Honore Pastries", "neighborhood": "RITTENHOUSE", "photo_count": 7, "review_count": 80, "stars": 4, "state": "PA" }]	No links
404	Not found	No links

Code	Description	Links
422	Validation Error	No links
Media type		
application/json		
Example Value Schema		
<pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }] }</pre>		

GET

/api/businesses/viewport Get Businesses In Viewport



Get businesses within a geographic viewport (bounding box) with optional filters.

Primary use case: Dynamic map loading - load only businesses visible in the current map view.

Viewport bounds (required):

- Define the geographic area to search within
- Automatically updates as user pans/zooms the map

Optional filters (combine for precise results):

- **state** : State code (e.g., 'TN', 'PA')
- **city** : City name (e.g., 'Nashville', 'Philadelphia')
- **category** : Category keyword (e.g., 'Restaurants', 'Coffee')
- **min_rating** : Minimum star rating (0-5)
- **is_open** : Business status (0=closed, 1=open)

Performance:

- Returns up to 5000 businesses per request
- Results ordered by rating and review count
- Much faster than loading all businesses and filtering client-side

Examples:

- Map viewport only: `?south=36.0&north=36.3&west=-87.0&east=-86.5`
- Restaurants in viewport: `?south=36.0&north=36.3&west=-87.0&east=-86.5&category=Restaurants`
- 4+ star open businesses: `?south=36.0&north=36.3&west=-87.0&east=-86.5&min_rating=4&is_open=1`
- Nashville only: `?south=36.0&north=36.3&west=-87.0&east=-86.5&city=Nashville&state=TN`

Parameters

Try it out

Name	Description
south * required number (query)	Southern latitude bound south <i>maximum: 90</i> <i>minimum: -90</i>
north * required number (query)	Northern latitude bound north <i>maximum: 90</i> <i>minimum: -90</i>
west * required number (query)	Western longitude bound west <i>maximum: 180</i> <i>minimum: -180</i>
east * required number (query)	Eastern longitude bound east <i>maximum: 180</i> <i>minimum: -180</i>
state string (string null) (query)	Filter by state code (e.g., 'TN', 'PA') state <i>minLength: 2</i> <i>maxLength: 2</i>
city string (string null) (query)	Filter by city name city <i>minLength: 1</i>
neighborhood string (string null) (query)	Filter by neighborhood name neighborhood <i>minLength: 1</i>
category string (string null) (query)	Filter by category (partial match) category <i>minLength: 1</i>
min_rating number (number null) (query)	Filter by minimum star rating (0-5) min_rating <i>maximum: 5</i> <i>minimum: 0</i>
is_open integer (integer null) (query)	Filter by open status (0=closed, 1=open) is_open <i>maximum: 1</i> <i>minimum: 0</i>

Name	Description
limit	Maximum number of businesses to return
integer	
(query)	Default value : 1000
	1000
	maximum: 5000 minimum: 1

Responses

Code	Description	Links
200	Successful Response	No links
	Media type application/json Controls Accept header. Example Value Schema [{ "attributes": { "Alcohol": "u'none'", "RestaurantsDelivery": "False", "WiFi": "u'free'" }, "business_id": "MTSW4McQd7CbVtyjqoe9mw", "categories": "Restaurants, Food, Bubble Tea, Coffee & Tea, Bakeries", "city": "Philadelphia", "hours": { "Monday": "7:0-20:0", "Tuesday": "7:0-20:0" }, "is_open": 1, "latitude": 39.9555052, "longitude": -75.1555641, "name": "St Honore Pastries", "neighborhood": "RITTENHOUSE", "photo_count": 7, "review_count": 80, "stars": 4, "state": "PA" }]	
404	Not found	No links
422	Validation Error	No links
	Media type application/json Example Value Schema	

Code	Description	Links
	<pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }] }</pre>	

GET `/api/businesses/search` Search Businesses



Search businesses using advanced fuzzy matching.

Features:

- **Multi-term search:** "Philadelphia Italian" finds Italian restaurants in Philadelphia
- **Fuzzy matching:** Handles typos ("Philadelfia" matches "Philadelphia")
- **Intelligent ranking:** Best matches first based on relevance, not just ratings
- **Cross-field search:** Searches name, city, state, and categories simultaneously

Search fields:

- Business name
- City
- State
- Categories

Examples:

- **a=nizza** → All nizza places
- **a=Philadelnhia** → All businesses in Philadelphia
- **a=Philadelnhia Italian** → Italian restaurants in Philadelphia
- **a=Philadelfia restaurant** → Restaurants in Philadelphia (typo corrected!)
- **a=PA coffee** → Coffee shops in Pennsylvania
- **q=cheesesteak** → Places serving cheesesteaks

How it works: Each word in your query contributes to finding the best matches. The more relevant the match, the higher it ranks.

Parameters

Try it out

Name	Description
q * required string (query)	Search query - supports multi-term and fuzzy matching q <i>minLength: 1</i>
skip integer (query)	Number of records to skip <i>Default value : 0</i>

Name	Description
	0 <i>minimum: 0</i>
limit integer (query)	Maximum number of results to return <i>Default value : 20</i> 20 <i>maximum: 100</i> <i>minimum: 1</i>

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema [{ "attributes": { "Alcohol": "u'none'", "RestaurantsDelivery": "False", "WiFi": "u'free'" }, "business_id": "MTSW4McQd7CbVtyjqoe9mw", "categories": "Restaurants, Food, Bubble Tea, Coffee & Tea, Bakeries", "city": "Philadelphia", "hours": { "Monday": "7:0-20:0", "Tuesday": "7:0-20:0" }, "is_open": 1, "latitude": 39.9555052, "longitude": -75.1555641, "name": "St Honore Pastries", "neighborhood": "RITTENHOUSE", "photo_count": 7, "review_count": 80, "stars": 4, "state": "PA" }]	No links
404	Not found	No links
422	Validation Error	No links
	Media type	

Code	Description	Links
------	-------------	-------

application/json

Example Value Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

GET	/api/businesses/{business_id}	Get Business	^
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Get a single business by ID.

Returns detailed business information including location, ratings, hours, and attributes.

Parameters

Try it out

Name	Description
------	-------------

business_id * required	Unique business identifier
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string
(path)

business_id

minLength: 1

Responses

Code	Description	Links
------	-------------	-------

200	Successful Response	No links
-----	---------------------	----------

Media type

application/json

Controls Accept header.

Example Value Schema

```
{
  "attributes": {
    "Alcohol": "u'none'",

```


Code	Description	Links
	<pre>"RestaurantsDelivery": "False", "WiFi": "u'free'" }, "business_id": "MTSW4McQd7CbVtyjqoe9mw", "categories": "Restaurants, Food, Bubble Tea, Coffee & Tea, Bakeries", "city": "Philadelphia", "hours": { "Monday": "7:0-20:0", "Tuesday": "7:0-20:0" }, "is_open": 1, "latitude": 39.9555052, "longitude": -75.1555641, "name": "St Honore Pastries", "neighborhood": "RITTENHOUSE", "photo_count": 7, "review_count": 80, "stars": 4, "state": "PA" }</pre>	
404	Not found	No links
422	Validation Error	No links
Media type		
application/json		
Example Value Schema		
<pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }] }</pre>		

locations



GET	/api/states	Get States	
Get list of all states that have businesses. Returns a list of 2-letter state codes (e.g., 'PA', 'TN', 'CA') for states that contain at least one business in the database. Use Case: Populate state dropdown/filter in the dashboard UI. Example Response: <pre>["AZ", "CA", "FL", "NV", "PA", "TN"]</pre>			

Parameters

Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>["string"]</pre>	No links

GET

/api/cities Get Cities



Get list of all cities in a specific state that have businesses.

Returns city names alphabetically sorted for the given state code.

Use Case: Populate city dropdown after user selects a state.

Parameters:

- state** : 2-letter state code (case-insensitive, converted to uppercase)

Example Request: `GET /cities?state=PA`

Example Response:

```
["Allentown", "Philadelphia", "Pittsburgh"]
```

Parameters

Try it out

Name	Description
state * required string (query)	State code (e.g., 'PA', 'TN') state minLength: 2 maxLength: 2

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema ["string"]	No links
422	Validation Error Media type application/json Example Value Schema { "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }] }	No links

GET /api/neighborhoods Get Neighborhoods



Get list of all neighborhoods in a specific city that have businesses.

Returns neighborhood names for the given city/state combination. Neighborhoods are assigned to businesses based on geographic boundaries.

Use Case: Populate neighborhood filter dropdown after user selects a city.

Parameters:

- **state** : 2-letter state code (case-insensitive)
- **city** : City name (case-sensitive match)

Example Request: `GET /neighborhoods?state=PA&city=Philadelphia`

Example Response:

`["Center City", "Fishtown", "Northern Liberties", "University City"]`

Parameters

Try it out

Name	Description
state * required string (query)	State code (e.g., 'PA', 'TN') state <i>minLength: 2</i> <i>maxLength: 2</i>
city * required string (query)	City name (e.g., 'Philadelphia', 'Nashville') city <i>minLength: 1</i>

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema ["string"]	No links
422	Validation Error Media type application/json Example Value Schema	No links

Code	Description	Links
	<pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }] }</pre>	

GET	/api/neighborhoods/boundaries	Get Neighborhood Boundaries	⤴
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Get GeoJSON boundaries for neighborhoods in a specific city.

Returns a GeoJSON FeatureCollection containing polygon boundaries for all neighborhoods in the specified city. Used for rendering neighborhood overlays on the map.

Use Case: Display neighborhood boundaries on the interactive map.

Parameters:

- **state** : 2-letter state code
- **city** : City name

Response Format: GeoJSON FeatureCollection

```
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "Feature",
      "properties": {"name": "Center City"},
      "geometry": {"type": "Polygon", "coordinates": [...]}
    }
  ]
}
```

Errors:

- **404** : Neighborhood boundaries not available for the specified city
- **500** : Error loading boundary file

Parameters

Try it out

Name	Description
state * required string (query)	State code (e.g., 'PA', 'TN')
	state minLength: 2 maxLength: 2

Name	Description
city * required string (query)	City name (e.g., 'Philadelphia', 'Nashville') city <i>minLength: 1</i>

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema "string"	No links
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }] }</pre>	No links

GET /api/cities/boundaries Get City Boundaries



Get GeoJSON boundary for an entire city.

Returns a GeoJSON Feature containing the polygon boundary for the specified city. Used for rendering city outline on the map and constraining the viewport.

Use Case: Display city boundary overlay and set initial map bounds.

Parameters:

- **state** : 2-letter state code
- **city** : City name

Response Format: GeoJSON object

```
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "Feature",
      "properties": {"name": "Philadelphia"},
      "geometry": {"type": "Polygon", "coordinates": [...]}
    }
  ]
}
```

Errors:

- **404** : City boundary not available
- **500** : Error loading boundary file

Parameters

Try it out

Name	Description
state * required string (query)	State code (e.g., 'PA', 'TN') state minLength: 2 maxLength: 2
city * required string (query)	City name (e.g., 'Philadelphia', 'Nashville') city minLength: 1

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema "string"	No links

Code	Description	Links
422	Validation Error	No links

Media type

application/json

Example Value

Schema

```
{  "detail": [    {      "loc": [        "string",        0      ],      "msg": "string",      "type": "string"    }  ]}
```

analytics

^

GET

/api/analytics/business/{business_id}/combined-timeline

Get Business Combined Timeline

^

Get combined ratings and sentiment timelines for a business with city and category comparisons.

Parameters

Try it out

Name	Description
business_id * required	Business identifier
string (path)	business_id
period	Time period for aggregation
string (query)	Default value : month month pattern: ^(day/week/month/year)\$
start_date	Start date filter (YYYY-MM-DD)
string (string null) (\$date) (query)	start_date
end_date	End date filter (YYYY-MM-DD)

Name	Description
<code>string (string null)</code> <code>(\$date)</code> <i>(query)</i>	<code>end_date</code>
<code>category</code> <code>string (string null)</code> <i>(query)</i>	Optional category for comparison (user-selected) <code>category</code>

Responses

Code	Description	Links
200	Successful Response	No links
	Media type application/json Controls Accept header. Example Value Schema <pre>{ "additionalProp1": {} }</pre>	
404	Not found	No links
422	Validation Error	No links
	Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }] }</pre>	

 $\wedge$

Get combined ratings and sentiment timelines for a city with optional category comparison.

Parameters

Try it out

Name	Description
state * required string (path)	State code (e.g., 'PA') state <i>minLength: 2</i> <i>maxLength: 2</i>
city * required string (path)	City name city <i>minLength: 1</i>
period string (query)	Time period for aggregation Default value : month month <i>pattern: ^(day/week/month/year)\$</i>
start_date string (string null)(\$date) (query)	Start date filter (YYYY-MM-DD) start_date
end_date string (string null)(\$date) (query)	End date filter (YYYY-MM-DD) end_date
category string (string null) (query)	Optional category for comparison category

Responses

Code	Description	Links
200	Successful Response Media type application/json	No links

Code	Description	Links
	Controls Accept header. Example Value Schema <pre>{ "additionalProp1": {}}</pre>	
404	Not found	No links
422	Validation Error	No links
	Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }]}</pre>	

GET

/api/analytics/category/{category}/combined-timeline

Get Category Combined Timeline

Get combined ratings and sentiment timelines for a category.

Parameters

Try it out

Name	Description
category * required string (path)	Category name category minLength: 1
period string (query)	Time period for aggregation Default value : month month pattern: ^(day/week/month/year)\$
start_date	Start date filter (YYYY-MM-DD)

Get combined ratings and sentiment timelines for a neighborhood with optional category comparison.

Reduces API calls by combining neighborhood ratings, neighborhood sentiment, and optional category data.

Parameters

Try it out

Name	Description
state * required string (path)	State code (e.g., 'PA') state <i>minLength: 2</i> <i>maxLength: 2</i>
city * required string (path)	City name city <i>minLength: 1</i>
neighborhood * required string (path)	Neighborhood name neighborhood <i>minLength: 1</i>
period string (query)	Time period for aggregation Default value : month month <i>pattern: ^(day/week/month/year)\$</i>
start_date string (string null) (\$date) (query)	Start date filter (YYYY-MM-DD) start_date
end_date string (string null) (\$date) (query)	End date filter (YYYY-MM-DD) end_date
category string (string null) (query)	Optional category for comparison category

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>{ "additionalProp1": {}}</pre>	No links
404	Not found	No links
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }]}</pre>	No links

GET

/api/analytics/competitive-snapshot

Get Competitive Snapshot

^

Get competitive positioning snapshot for market analysis.

Returns all businesses in the specified market (city/category) with pre-calculated statistics for competitive positioning visualization (scatter plots, quadrant analysis).

Use Case: Competitive market positioning visualization (RQ2)

- Show how a business compares to competitors in rating vs review volume space
- Identify market leaders, hidden gems, and at-risk businesses
- Provide actionable insights about competitive position

Filters (at least one recommended):

- **city** + **state** : All businesses in a city (e.g., Philadelphia, PA)
- **category** : All businesses in a category (e.g., Restaurants)
- Combine filters for precision (e.g., Restaurants in Philadelphia)

Performance: Returns up to 5000 businesses with pre-calculated market statistics

Example Response:

```
{
  "businesses": [
    {
      "business_id": "abc123",
      "name": "Joe's Pizza",
      "stars": 4.5,
      "review_count": 234,
      "city": "Philadelphia",
      "state": "PA",
      "categories": "Restaurants, Pizza, Italian",
      "is_open": 1
    }
  ],
  "statistics": {
    "avg_rating": 3.8,
    "median_review_count": 45,
    "total_businesses": 1234
  },
  "selected_business": {...},
  "filters": {
    "city": "Philadelphia",
    "state": "PA",
    "category": "Restaurants"
  }
}
```

Parameters

Try it out

Name	Description
city	Filter by city name
string (string null) (query)	city
state	Filter by state code (e.g., 'PA')
string (string null) (query)	state <i>minLength: 2</i> <i>maxLength: 2</i>
neighborhood	Filter by neighborhood name
string (string null) (query)	neighborhood
category	Filter by category (partial match)
string (string null) (query)	category
business_id	Specific business to highlight
string (string null) (query)	business_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>{ "additionalProp1": {}}</pre>	No links
404	Not found	No links
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }]}</pre>	No links

GET

/api/analytics/business/{business_id}/forecast

Get Business Forecast

^

Generate rating and sentiment forecasts for a business.

Uses ARIMA modeling when sufficient data points exist (6+), falls back to mean-based projection for sparse data.

Response:

- rating forecast : Predicted ratings with 80% confidence bands
- sentiment forecast : Predicted sentiment scores with confidence bands
- model_type : 'arima' or 'fallback' indicating method used

Parameters

Try it out

Name	Description
business_id * required string (path)	Business identifier business_id
periods integer (query)	Number of periods to forecast Default value : 4 4 maximum: 12 minimum: 1
period_type string (query)	Period type for forecast Default value : month month pattern: ^(month/year)\$

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>{ "additionalProp1": {} }</pre>	No links
404	Not found	No links
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [</pre>	No links

Code	Description	Links
	<pre>{ "loc": ["string", 0], "msg": "string", "type": "string" }</pre>	

GET

/api/analytics/business/{business_id}/period-issues

Get Period Issues



Parameters

Try it out

Name	Description
business_id * required string (path)	Business identifier business_id
start_date * required string(\$date) (query)	Start date (YYYY-MM-DD) start_date
end_date * required string(\$date) (query)	End date (YYYY-MM-DD) end_date

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>{ "additionalProp1": {} }</pre>	No links
404	Not found	No links

Code	Description	Links
422	Validation Error Media type application/json Example ValueSchema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }]}</pre>	No links

health

^

GET	/ Root	▼
GET	/health Health Check	▼

Schemas	^
BusinessDTO > Expand all object HTTPValidationError > Expand all object ValidationError > Expand all object	